Roots of d-matching polynomials and the spectrum of the universal cover: a closed form at first Betti number one, and a refuted conjecture

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Abstract

Hall, Puder and Sawin place the roots of the d-matching polynomial $\mu_{d,G}$ in the interval $[-\rho, \rho]$, ρ the spectral radius of the universal cover T. We conjectured that they lie in $\mathrm{spec}(T)$ itself, a proper subset of that interval whenever T has a spectral gap: no root ever enters a gap. This strengthens their theorem and is the matching-polynomial form of their Question 6.3.

We prove it for every connected G of first Betti number one and every d, by giving $\mu_{d,G}$ in closed form.

In general it is false. The counterexample is due to Chris Hall [12]: a bipartite graph on 41 vertices with $\mu_G = x^{21}(x^4 - 11x^2 + 25)^4(x^2 - 5)(x^2 - 11)$, where $\sqrt{5}$ lies in an internal gap of $\mathrm{spec}(T)$. We give the mechanism. A separator of size k met by p identical branches forces $A^{p-k} \mid \mu_G$, where A is the branch's own matching polynomial (Lemma 13); Hall's $\sqrt{5}$ is the top matching root of the star $K_{1,5}$, and a star's leaves have degree one in the divisor however large their degree is in G. Hypotheses on G therefore cannot reach the offending root. We show this is not a figure of speech by refuting both repairs that have been proposed: minimum degree three fails on 14 vertices, with $\mu_G = x^4(x^2 - 3)(x^8 - 24x^6 + 171x^4 - 396x^2 + 270)$ and $\sqrt{3}$ outside $\mathrm{spec}(T)$, and 3-connectivity fails on 23. Any repair must constrain the divisor rather than the graph.

Everything asserted is machine-checked in Lean 4/Mathlib unless labelled otherwise. Supplementary material is in [17]; the biregular case, where the conjecture survives, is [16].

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Keywords. matching polynomial, d-matching polynomial, universal cover, spectral gap, graph covering, interlacing families, Lean 4, formalization.

Code and data availability. All computations, the Lean 4/Mathlib sources, and the scripts backing every numerical statement are in the `rama-notes` repository. Each numerical claim names the script that produced it; every named script is executed and its output compared against a recorded baseline by `code/cite_check.py`, and every Lean declaration cited is checked for its axiom base by `code/audit_papers.py`.

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Status conventions. Claims here are of three kinds, and the text says which each time rather than leaving it to the reader. A result described as *formalized*, or cited by a Lean declaration name, is machine-checked in Lean 4/Mathlib: the sources are in `RamaLean/` of the accompanying repository, carry no `sorry`, and rest only on the standard axioms `propext`, `Classical.choice` and `Quot.sound`. A claim described as *measured*, *verified to* a stated tolerance, or *checked numerically* is computational evidence and is not a proof; the script producing it is named, and every named script is run and its output compared against a recorded baseline (`code/cite_check.py`). A claim called a *conjecture* is unproved, and is called that even where the evidence is strong.

1 Introduction and statement

The expected characteristic polynomial of a random lift is the object at the heart of the interlacing-families method of Marcus–Spielman–Srivastava [11], who used the $r = 2$ (signing) case to prove that bipartite Ramanujan graphs exist in every degree; Hall–Puder–Sawin [2] extended the framework to arbitrary coverings. We evaluate it in closed form for the cycle.

A *permutation r-lift* of a graph G replaces each vertex by r copies and each edge $\{u, v\}$ by the perfect matching $u_i \sim v_{\sigma(i)}$ of a permutation $\sigma \in S_r$ chosen independently and uniformly per edge. For the cycle C_n, gauge-fixing a spanning path reduces the lift to a single permutation $\sigma \in S_r$ on the closing edge, and the lift decomposes into disjoint cycles: each ℓ-cycle of σ contributes a cycle $C_{n\ell}$ (fixed points contribute copies of C_n). Since $\chi_{C_m}(x) = \det(xI - A_{C_m}) = 2T_m(x/2) - 2$, the expected characteristic polynomial is the expected cycle weight

$$\Phi_{n,r}(x) \;=\; \mathbb{E}_{\sigma\in S_r} \prod_{\ell\,\in\,\mathrm{cycles}(\sigma)} \big(2T_{n\ell}(x/2) - 2\big). \tag{1}$$

Theorem 1. *For all $n \geq 1$, $r \geq 1$, with $Y = T_n(x/2)$,*

$$\Phi_{n,r}(x) \;=\; U_r(Y) - 2U_{r-1}(Y) + U_{r-2}(Y) \;=\; \underbrace{\big(2T_n(x/2) - 2\big)}_{\chi_{C_n}(x)} \cdot \underbrace{U_{r-1}\big(T_n(x/2)\big)}_{\Psi_r(x)}.$$

The factored form identifies the quotient Ψ_r, by the theorem of Hall–Puder–Sawin [2] equating the expected new-eigenvalue characteristic polynomial with the d-matching polynomial ($d = r - 1$), as

$$\mu_{d,C_n}(x) \;=\; U_d\big(T_n(x/2)\big),$$

which is equivalent (through $\mathcal{P}_d(2t) = U_d(t)$, $\mathcal{C}_n(2t) = 2T_n(t)$ and the composition identity $U_{dn+n-1} = U_d(T_n)\,U_{n-1}$) to *Hall's conjecture* $\mathcal{C}_{n,d} = \mathcal{P}_d(\mathcal{C}_n)$, proved combinatorially in [3]. That preprint has remained unrefereed since 2018 and carries no journal reference or DOI as of August 2026. The reason is not mathematical. Chris Hall, who posed the conjecture, has told us (personal communication, August 2026) that Andrew Herring was his first doctoral student; Hall gave him the conjecture, Herring presented it as a problem at a workshop, and the group of [3] solved it there. They submitted the result, met a couple of rejections, and abandoned the attempt for want of motivation rather than for any defect. Hall regards the proof as correct, and so do we. We record the episode because a reader weighing what is established about this identity should know both that the combinatorial proof stands and that the present derivation is, as far as we can tell, the first refereed or machine-checked one. Our route is different and elementary; we make no use of matching-polynomial combinatorics.

Corollary 2. $\sum_{d\geq 0} \mu_{d,C_n}(x)\, z^d = \dfrac{1}{1-2T_n(x/2)\,z+z^2}$, *and* $\sum_{r\geq 0} \Phi_{n,r}(x)\, z^r = \dfrac{(1-z)^2}{1-2T_n(x/2)\,z+z^2}$.

Corollary 3 (real-rootedness)**.** *All roots of* $\Phi_{n,r}$ *lie in* $[-2,2]$*: with* $x = 2\cos\theta$*,* χ_{C_n} *vanishes at* $\theta = 2\pi m/n$ *and* $\Psi_r = \sin(rn\theta)/\sin(n\theta)$ *vanishes at* $\theta = m\pi/(rn)$ *with* $r \nmid m$ *(when* $r \mid m$ *the denominator vanishes too and* $\Psi_r = \pm r \neq 0$*); this leaves exactly* $n(r-1)$ *roots, matching* $\deg \Psi_r$*.*

Corollary 4 (Fibonacci–Lucas evaluation)**.** $\Phi_{n,r}(3) = (L_{2n}-2)\, F_{2nr}/F_{2n}$*, since* $T_n(3/2) = L_{2n}/2$ *and* $U_{r-1}(L_{2n}/2) = F_{2nr}/F_{2n}$*.*

Corollary 5 (parity)**.** $\Psi_r(-x) = (-1)^{n(r-1)}\Psi_r(x)$*: the quotient is supported on exponents of a single parity, like a matching polynomial. (For bipartite* C_n *it is an even polynomial for every* r*; at* $r = 2$ *it is the classical matching polynomial* $2T_n(x/2)$ *of* C_n*, consistent with Godsil–Gutman [1].)*

What this paper does

The closed form is the starting point rather than the subject. Read as a statement about where the roots of $\mu_{d,G}$ sit, it says that for the cycle they land exactly in $\operatorname{spec}(T)$, and not merely in the interval $[-\rho,\rho]$ that Hall–Puder–Sawin guarantee. The two differ whenever T has a spectral gap, and whether the stronger statement holds in general is the question here.

§3 proves it for every graph of first Betti number one and every d. §4 states it as a conjecture for all graphs. §5 is the counterexample, due to Chris Hall, together with the reason no local repair of it can work: by the Divisibility Lemma of §5.4 the branch polynomial divides μ_G across a separator, so the offending root belongs to the divisor, and hypotheses on G cannot see it. Minimum degree three fails on 14 vertices and 3-connectivity on 23. Supplementary material, including two families in which the conjecture is provable and the record of the searches that missed the counterexample, is in [17]; the biregular case, which survives the refutation, is [16].

2 Proof

Write $f(\ell) = 2T_\ell(Y) - 2$ with $Y = T_n(x/2)$; by the composition identity $T_{n\ell} = T_\ell \circ T_n$ this is the per-cycle factor in (1).

Step 1 (exponential formula). For any commutative ring R and $f : \mathbb{N} \to R$, the S_r-average $\Phi_r = \mathbb{E}_\sigma \prod_{\ell\in\text{cycles}(\sigma)} f(\ell)$ satisfies

$$r\,\Phi_r = \sum_{k=1}^{r} f(k)\,\Phi_{r-k}, \qquad \Phi_0 = 1, \tag{2}$$

the coefficient form of $\sum_r \Phi_r z^r = \exp\bigl(\sum_\ell f(\ell) z^\ell/\ell\bigr)$. (Proof: group σ by cycle type λ with Cauchy's count $r!/z_\lambda$, mark a part, and re-index; see the Lean file `Paper2ExpFormula.lean` for the fully detailed version.)

Step 2 (Chebyshev generating function). With $f(\ell) = 2T_\ell(Y) - 2$,

$$\sum_{\ell\geq 1} f(\ell)\frac{z^\ell}{\ell} = -\log\bigl(1-2Yz+z^2\bigr) + 2\log(1-z) \quad\Longrightarrow\quad \sum_{r\geq 0}\Phi_r z^r = \frac{(1-z)^2}{1-2Yz+z^2}.$$

Extracting coefficients against $\sum U_r(Y)z^r = (1-2Yz+z^2)^{-1}$ gives $\Phi_r = U_r - 2U_{r-1} + U_{r-2}$, and the three-term recurrence $U_r + U_{r-2} = 2YU_{r-1}$ collapses this to $(2Y-2)\,U_{r-1}(Y)$. $\square$

In the formalization the analytic Step 2 is replaced by an equivalent finite argument: the closed form is shown to satisfy (2) directly (a second-difference induction), and (2) with $\Phi_0 = 1$ determines Φ over a characteristic-zero domain.

3 Graphs of first Betti number one

The proof above uses cycles in exactly one place: gauge-fixing a spanning path leaves a *single* permutation, because C_n has $|E| - |V| + 1 = 1$. The argument therefore applies verbatim to any connected G with $b_1(G) = |E| - |V| + 1 = 1$, a cycle with trees attached, and we record what it gives.

By [2] the expected characteristic polynomial of a random r-lift of *any* graph factors as $\chi_G \cdot \mu_{r-1,G}$; the content below is therefore not that factorization but a closed form for the factor $\mu_{d,G}$ itself. Question 6.5 of [2] asks which parts of the theory of the matching polynomial $\mu_{1,G}$ generalize to $\mu_{d,G}$; the following answers it for $b_1(G) = 1$.

Let C be the unique cycle of G and let $G - V(C)$ be the forest obtained by deleting its vertices. Write μ_H for the ordinary matching polynomial of a graph H, with $\mu_\emptyset = 1$. Define

$$V_0 = 1, \qquad V_1 = \mu_G, \qquad V_d = \mu_G\, V_{d-1} - \mu_{G-V(C)}^2\, V_{d-2}. \tag{3}$$

Theorem 6. *For every connected G with $b_1(G) = 1$ and every $d \geq 0$, $\mu_{d,G} = V_d$.*

Theorem 6 is proved below, through the twisted characteristic polynomial; it is not formalized.

For $G = C_n$ the deleted forest is empty, so $\mu_{G-V(C)} = 1$ and (3) is the recurrence for $U_d(\mu_{C_n}/2) = U_d(T_n(x/2))$: Theorem 1 is the case $\mu_{G-V(C)} = 1$. In general the attached trees enter only through $\mu_{G-V(C)}$; for the triangle with one pendant edge, $\mu_G = x^4 - 4x^2 + 1$ and $\mu_{G-V(C)} = x$.

The proof goes through the twisted characteristic polynomial. Let $e = uv$ be the unique non-tree edge, let $A_G(z)$ be the adjacency matrix of G with the (u,v) entry replaced by z and the (v,u) entry by z^{-1}, and set $F(x,z) = \det(xI - A_G(z))$.

Lemma 7. *$F(x,z) = \mu_G(x) - \mu_{G-V(C)}(x)\,(z+z^{-1})$.*

Proof. By the Sachs coefficient theorem $\det(xI - A)$ expands over elementary subgraphs, that is disjoint unions of edges and cycles. Since a permutation uses each of the two twisted entries at most once and $A_G(z)^\mathsf{T} = A_G(z^{-1})$, the determinant is of the form $P(x) + Q(x)(z+z^{-1})$. Replacing z by ± 1 and averaging kills exactly the elementary subgraphs whose cycle traverses

e; as G has the single cycle C, those are the subgraphs containing C, so $P = \mu_G$, and the surviving terms contribute $Q = -\mu_{G-V(C)}$, a cycle component carrying the Sachs weight -2. $\square$

Proof of Theorem 6. Gauge-fix a spanning tree: the r-lift is determined by one permutation $\sigma \in S_r$ on e, and its components are indexed by the cycles of σ, the component of an ℓ-cycle being the ℓ-fold cyclic cover G_ℓ. Decomposing the $\mathbb{Z}/\ell$ action into characters gives the classical factorization $\chi_{G_\ell}(x) = \prod_{j=0}^{\ell-1} F(x, \omega^j)$ with $\omega = e^{2\pi i/\ell}$ [9, 10]. With P, Q as in Lemma 7 and $y = -P/(2Q)$,

$$\chi_{G_\ell} = \prod_{j=0}^{\ell-1}\Big(P + 2Q\cos\tfrac{2\pi j}{\ell}\Big) = (-Q)^\ell\big(2T_\ell(y) - 2\big),$$

so by the exponential formula (2), with $w = -Qz$,

$$\sum_{r\geq 0}\Phi_{G,r}z^r = \exp\Big(\sum_{\ell\geq 1}\big(2T_\ell(y) - 2\big)\frac{w^\ell}{\ell}\Big) = \frac{(1-w)^2}{1 - 2yw + w^2},$$

and extracting coefficients as in Section 2 gives $\Phi_{G,r} = \chi_G \cdot (-Q)^{r-1}U_{r-1}(y)$. Comparing with $\Phi_{G,r} = \chi_G \cdot \mu_{r-1,G}$ [2] yields $\mu_{d,G} = (-Q)^d U_d(y)$, which satisfies (3). $\square$

Since $U_d(y) = 2^d\prod_{k=1}^d(y - \cos\frac{k\pi}{d+1})$, the theorem can be restated as a product, $\mu_{d,G}(x) = \prod_{k=1}^d F\big(x, e^{\,ik\pi/(d+1)}\big)$, whence:

Corollary 8. *For $b_1(G) = 1$, $\mu_{d,G}$ is a product of d real-rooted polynomials, and its roots are exactly the eigenvalues of the Hermitian matrices $A_G\big(e^{\,ik\pi/(d+1)}\big)$, $1 \leq k \leq d$.*

This is the real-rootedness theorem of [2] for this family, with the roots identified. The localization itself, that a vanishing V_d forces $|\mu_G| \leq 2|\mu_{G-V(C)}|$, so that a root can never sit in a spectral gap, is machine-checked in Lean 4 as `cV_root_mem_band`, resting on `cU_ne_zero_of_one_le_abs`: the Chebyshev polynomial U_d has no root of absolute value at least one.

The roots can be located much more precisely than that. Since $b_1(G) = 1$ the group $\pi_1(G)$ is $\mathbb{Z}$, so the universal cover T of G *is* the $\mathbb{Z}$-cover, and T is a periodic decorated chain whose adjacency operator decomposes by Floquet–Bloch into the fibres $A_G(e^{i\theta})$. Hence

$$\mathrm{spec}(T) = \bigcup_\theta \mathrm{spec}\,A_G(e^{i\theta}) = \Big\{x :\ \exists\, s \in [-1,1],\ \mu_G(x) = 2s\,\mu_{G-V(C)}(x)\Big\},$$

a finite union of bands, together with a flat band at each common zero of μ_G and $\mu_{G-V(C)}$, where $F(x,\cdot)$ vanishes identically.

Corollary 9. *For $b_1(G) = 1$ the roots of $\mu_{d,G}$ are exactly the solutions of $\mu_G(x) = 2\cos\big(\frac{k\pi}{d+1}\big)\mu_{G-V(C)}(x)$ for $1 \leq k \leq d$. In particular every root of every $\mu_{d,G}$ lies in* $\mathrm{spec}(T)$, *and as $d \to \infty$ the roots equidistribute on* $\mathrm{spec}(T)$ *with respect to its density of states.*

Proof. Immediate from Theorem 6 in the form $\mu_{d,G} = \prod_k F(x, e^{ik\pi/(d+1)})$ and $F(x, e^{i\theta}) = \mu_G - 2\cos\theta\,\mu_{G-V(C)}$. The equidistribution is the arcsine law for $\cos\big(k\pi/(d+1)\big)$ transported through the fibre map. $\square$

Hall–Puder–Sawin place the roots of $\mu_{d,G}$ in the Ramanujan *interval* $[-\rho, \rho]$, ρ being the spectral radius of T. Corollary 9 places them in $\operatorname{spec}(T)$, which is a union of bands and is a proper subset of that interval whenever T has a spectral gap: no root of any $\mu_{d,G}$ can ever enter a gap. The two statements coincide exactly when the band structure is trivial, and that is precisely the case of a cycle, where $\mu_{G-V(C)} = 1$ and $\operatorname{spec}(T) = [-2, 2]$ is the whole interval. Attaching even a single pendant edge opens gaps, and they are not small (computed band structures, proportion of $[-\rho, \rho]$ that is a gap):

G	ρ	gap proportion
C_n	2	0%
triangle + pendant	2.170	36.5%
C_4 + pendant	2.136	45.9%
triangle + P_2	2.214	51.3%
$C_4 + P_3$	2.189	54.1%
triangle + $K_{1,3}$	2.514	65.9%

So the cycle is not merely the easiest case of Theorem 6: it is the degenerate one, the unique member of the family whose spectrum has no gaps and for which the Ramanujan interval is attained. Any cycle with a tree attached exhibits a band structure that no cycle can display.

The boundary is sharp. For $b_1(G) = b$ the lift is determined by b permutations, that is by an action of the free group F_b on $[r]$, whose orbits are the components of the lift. The exponential formula still applies, in the form $N_r = \sum_\ell \binom{r-1}{\ell-1} C_\ell N_{r-\ell}$ with $N_r = (r!)^b \Phi_{G,r}$ and C_ℓ the sum of χ over the transitive b-tuples on $[\ell]$; we have verified this for $b \le 3$. What fails is the atom. For $b = 1$ every transitive tuple gives the *same* cover G_ℓ, so $C_\ell = (\ell-1)!\,\chi_{G_\ell}$: one graph per ℓ, with χ_{G_ℓ} Chebyshev in ℓ. For $b \ge 2$ the transitive tuples number $(\ell-1)!$ times the count of index-ℓ subgroups of F_b and spread over many non-isomorphic covers with distinct spectra, so C_ℓ carries no such structure. Accordingly Theorem 6 is not merely unproved for $b \ge 2$: it is false there already at $d = 1$, as we checked for the theta graph, the bowtie, K_4, $K_{2,3}$ and two triangles joined by an edge.

Theorem 6 was verified by exact integer computation against the definition of $\mu_{d,G}$ as an average of matching polynomials over all d-covers, and against brute-forced lift averages, on thirteen named unicyclic graphs for $r \le 6$ and on a random sample of thirty-four for $r \le 5$; Lemma 7 was checked on eleven.

4 A conjecture for all graphs

Corollary 9 is proved by Floquet theory, which is available only because $\pi_1(G) = \mathbb{Z}$. Its *statement*, however, makes sense for every graph, and we know one case of it: at $b_1(G) = 1$ it is Corollary 9, for every d. We therefore propose:

Conjecture 10. *For every finite graph G and every $d \ge 1$, all roots of $\mu_{d,G}$ lie in* $\operatorname{spec}(T)$, *the spectrum of the universal cover of G.*

Theorem 11 (Hall [12])**.** *Conjecture 10 is false, already at $d = 1$ and already for simple connected loopless bipartite graphs.*

The counterexample and its certificate are entirely due to Chris Hall. We record it here because it settles the central question of this paper, and because everything else in the note has to be read in its light. §5 states his construction, our independent verification of it, and what we can add.

The biregular case of Conjecture 10 at $d = 1$ is not new. It is Problem 1 of Song, Fan and Miao [5], posed in 2023: they ask whether the matching polynomial of the incidence graph B_H of a (d, r)-regular hypergraph has no root μ with $0 < |\mu| < |\sqrt{d-1} - \sqrt{r-1}|$, and every (d, r)-biregular bipartite graph is such an incidence graph. Their Theorem 1.8 attaches a consequence: an affirmative answer would give every (d, r)-regular hypergraph a left-sided Ramanujan 2-covering. The only bound in that direction we know of is their Lemma 5.5, which goes the other way, $0 < \mu_\tau(B_H) \leq \max\{\sqrt{d}, \sqrt{r}\}$. We are not aware of any proof, partial result or counterexample. Conjecture 10 extends their problem in two directions, to arbitrary graphs and to all d.

Question 6.3 of [2] asks the corresponding question for graph eigenvalues, and leaves it open: it distinguishes graphs whose nontrivial spectrum lies in the *interval* $[-\rho, \rho]$ from those whose nontrivial spectrum lies in $\mathrm{spec}(T)$, observes that the two notions separate exactly when T has a gap, and gives the biregular tree as the example. Conjecture 10 is the matching-polynomial form of that distinction.

The conjecture strengthens the root-localization theorem of [2], which places the roots in the interval $[-\rho, \rho]$ with ρ the spectral radius of T; the conjecture places them in $\mathrm{spec}(T)$ itself. The two agree whenever T has no spectral gap, in particular for regular G, where T is a regular tree and $\mathrm{spec}(T) = [-2\sqrt{q}, 2\sqrt{q}]$ is an interval. They differ exactly when T has a gap, and the smallest examples are biregular: for $K_{2,3}$ the universal cover is the $(3, 2)$-biregular tree, with spectrum $\{0\} \cup \pm[\sqrt{2}-1, \sqrt{2}+1]$ and hence a gap $(0, \sqrt{2}-1)$ inside the Ramanujan interval. We have tested the conjecture on the biregular family, where the universal cover is an (a, b)-biregular tree with spectrum $\{0\} \cup \pm[\,|s-t|, s+t\,]$, $s = \sqrt{a-1}$, $t = \sqrt{b-1}$, so the gap is $(0, |s-t|)$. In each case $\mu_{d,G}$ was computed by averaging matching polynomials over all d-covers. No root ever lies in the gap; we also record the smallest nonzero $|\text{root}|$ as a multiple of the gap edge, which measures how close the bound comes to being attained.

G	b_1	type	gap edge	d	min $\lvert\text{root}\rvert$/gap edge
$K_{2,3}$	2	$(3,2)$	0.4142	$1,2,3,4$	2.72, 2.19, 1.94, 1.78
$K_{2,4}$	3	$(4,2)$	0.7321	$1,2,3$	1.93, 1.65, 1.52
$K_{2,5}$	4	$(5,2)$	1.0000	$1,2,3$	1.66, 1.46, 1.37
$K_{3,4}$	6	$(4,3)$	0.3178	$1,2$	3.04, 2.31
$K_{3,5}$	8	$(5,3)$	0.5858	$1,2$	2.10, 1.73
subdivision of K_4	3	$(3,2)$	0.4142	$1,2$	1.97, 1.67

All of these have $b_1 \geq 2$, so none is covered by Corollary 9. It is worth recording that even $d = 1$ is not formal. There $\mu_{1,G}$ is the ordinary matching polynomial, whose roots are eigenvalues of the path tree of G (Godsil), and the path tree embeds in T; but eigenvalues of a finite subtree of T need not lie in $\mathrm{spec}(T)$ when T has a gap. For the $(3, 2)$-biregular tree, whose gap is $(0, \sqrt{2}-1)$, the path P_8 is a subtree with eigenvalue $2\cos(4\pi/9) = 0.3473$ inside that gap. So the $d = 1$ case does not follow from the path-tree argument. It is nevertheless a theorem for an infinite family.

5 Hall's counterexample

Everything in this section is due to Chris Hall, who constructed the graph, found the certificate, and wrote the verification note [12]. Our contribution is the independent check, the formalization of the algebra. The two-parameter family containing his graph is in [17].

5.1 The graph and its matching polynomial

Let G consist of a central vertex c; for $1 \le i \le 5$ a copy of $K_{2,5}$ with degree-five vertices v_i, w_i and middle vertices $u_{i,1}, \dots, u_{i,5}$; a pendant leaf ℓ_i at each w_i; and the edges cv_i. It is simple, connected, loopless and bipartite, with 41 vertices, 60 edges and $b_1(G) = 20$.

Writing H for one branch after deleting c, the vertex-deletion recurrence gives $\mu_{H-v} = x^5(x^2-6)$ and $\mu_H = x^4(x^4 - 11x^2 + 25)$, and then

$$\mu_G(x) = \mu_H(x)^4 \big(x\mu_H(x) - 5\mu_{H-v}(x)\big) = x^{21}(x^4 - 11x^2 + 25)^4 (x^2 - 5)(x^2 - 11), \quad (4)$$

so $\sqrt{5}$ is a simple root. Identity (4), the simplicity of the root, and the value of the cofactor are machine-checked in `RamaLean/HallCounterexample.lean` as an identity in $\mathbb{Z}[X]$ (`muG_eq`, `muG_root_sqrt5`, `muG_root_simple`).

5.2 The spectral exclusion

Angel, Friedman and Hoory [13] characterise invertibility of $A_T - \lambda I$ by the existence of a finite nonzero *ratio system*: an assignment r_e to directed edges with $\lambda = 1/r_e + \sum_{e \to f} r_f$, whose decay matrix $\mathcal{R}_{e,f} = |r_f|^2$ on non-backtracking pairs has spectral radius below one. Hall exhibits such a system at $\lambda = \sqrt{5}$ with eight edge types and ratios in $\mathbb{Q}(\sqrt{5}, \sqrt{41})$. Its decay matrix has 120 states; the ten states at the leaves are transient, and the recurrent block of 110 states reduces by symmetry to a 6×6 orbit quotient K for which

$$x = (20, 121, 47, 33, 28, 16)^{\mathsf{T}} \quad \text{satisfies} \quad Kx < x$$

componentwise, so $\rho(K) < 1$ by Perron–Frobenius. Hence $\sqrt{5} \notin \operatorname{spec}(A_T)$, and with (4) this proves Theorem 11. Since G is bipartite the same holds at $-\sqrt{5}$, and Hall–Puder–Sawin place $\pm\sqrt{5}$ inside $[-\rho(T), \rho(T)]$, so these are roots in a genuine internal gap.

The eight ratio identities, the elementary bounds $32/5 < \sqrt{41} < 13/2$, the strict positivity of all six components of $x - Kx$, and a Collatz–Wielandt lemma (a nonnegative matrix with a positive strictly subinvariant vector has spectral radius below one) are machine-checked in `RamaLean/HallCounterexample.lean`. The argument common to all three counterexamples of this paper is isolated once in `RamaLean/RatioCertificate.lean`: a ratio system with its decay matrix (`decay`, `decay_nonneg`), the Collatz–Wielandt step in the strict form the certificate needs (`decay_lt_one`), and the assembly with Angel–Friedman–Hoory as an explicit hypothesis (`spec_excluded`). The three instances differ only in the numbers supplied. A schema rather than closed-form algebra is the right level here: Hall's ratios lie in $\mathbb{Q}(\sqrt{5}, \sqrt{41})$, but that is a feature of his graph, and for the 31-vertex one the ratios admit no minimal polynomial of small degree and moderate height. The theorem of [13] is not in Mathlib and enters as an explicit hypothesis, in the shape it is consumed.

We verified the certificate independently, reconstructing every object from the graph rather than copying the displayed matrices: the matching numbers

$(1, 60, 1585, 24260, 238105, 1564976, 6973855, 20805500, 39784375, 44062500, 21484375)$; the ratio equations on all 120 directed edges; the strongly connected components of the decay digraph; the orbit quotient rebuilt from follower counts; and the decay rate by two independent routes, the componentwise certificate and the characteristic polynomial $t^6 - \frac{201+19s}{800}t^2 + \frac{21s-241}{250}$ with $s = \sqrt{41}$. That reconstruction is `code/hall_certificate.py`, which rebuilds each object from the construction and copies no displayed matrix: the matching numbers agree term for term and $\mu_G(\sqrt{5})$ vanishes to 10^{-45}; the ratio system reduces to eight unknowns by symmetry and is solved exactly in $\mathbb{Q}(\sqrt{5}, \sqrt{41})$, its solution satisfying all 120 equations to 4×10^{-16}; the ten transient states are exactly the leaf-incident directed edges and the recurrent block has 110; and the six components of $x - Kx$ reproduce the exact expressions carried by `RamaLean/HallCounterexample.lean` to 3×10^{-14}, with $\rho(K) = 0.9636233789$. Three of the eight ratios are negative, which is why a positive cavity iteration does not find the system. As a physical check, the non-backtracking cavity Green function at $\sqrt{5} + i\eta$ converges to Hall's real ratios as $\eta \downarrow 0$, with density of states vanishing linearly in η.

5.3 The mechanism is a separation

Every counterexample is built on a separation. If a vertex v separates G into p isomorphic branches H, the deletion recurrence gives

$$\mu_G = \mu_H^{p-1}\big(x\mu_H - p\,\mu_{H-v}\big), \tag{5}$$

and the root is placed by tuning p and H; this is `CutVertexMechanism.branch_factor`, with `root_of_branch_factor` transferring a root of the last factor to μ_G. Hall's graph and the graphs of [17] are of this shape.

A cut vertex is only the smallest separation. Two hubs u, v with p identical branches attached to both leave no cut vertex, since removing either hub leaves the graph joined through the other, while $\{u, v\}$ separates it into p pieces. Writing $A = \mu_B$ for the branch, A_1, A_2 for the branch with one anchor deleted and A_{12} for both, deletion at u and then at v gives

$$\mu_G = A^{p-2}\big(x^2A^2 - px(A_1 + A_2)A + pA_{12}A + p(p-1)A_1A_2\big), \tag{6}$$

which is `SeparationOrder.two_cut_factor`, with `root_of_two_cut`.

The bracket in (6) is *quadratic* in p where the bracket in (5) is linear. Read as polynomials in the branch count the two have degrees one and two, with leading coefficient A_1A_2 in the second case (`cutBracket_natDegree`, `twoCutBracket_natDegree`, `freedom_grows`). Excluding cut vertices does not remove the engine; it upgrades it.

Proposition 12. *There are 2-connected counterexamples to Conjecture 10.*

Let $B_{q,T}$ be $K_{2,q}$ with a path of T further vertices appended to one of its two degree-q vertices; attach the other degree-q vertex to u and the end of the path to v, and take p copies.

branch	p	n	$\kappa(G)$	θ	defect	gap width
$B_{4,2}$	7	58	2	2.137920	0.003574	0.14758
$B_{5,2}$	6	56	2	2.287307	0.005624	0.06636
$B_{5,3}$	6	62	2	2.317609	0.008332	0.09876

All three are simple, bipartite, of minimum degree two and maximum degree at most seven, and of vertex connectivity exactly two, computed by Menger as a unit-capacity maximum flow over every non-adjacent pair. Each θ is an algebraic number of degree 12 or 14 isolated from the exact integer bracket, and (6) is checked against a brute-force matching polynomial. That $\theta \notin \mathrm{spec}(T)$ is confirmed twice: the Angel–Friedman–Hoory decay rate at 40 digits is 0.943 to 0.959, and the density of states of the universal cover, from the complex cavity equations, falls by a factor of ten per decade of η at θ while an in-band control holds constant to four figures (`code/twocut.py`).

Equation (6) is the case $k = 2$ of a general expansion. With k hubs and p branches attached to all of them,

$$\mu_G = \sum_{S \subseteq [k]} (-1)^{|S|} x^{k-|S|} \sum_{P \vdash S} (p)_{|P|} \Big(\prod_{J \in P} A_J \Big) A^{p-|P|}, \tag{7}$$

with A_J equal to μ_B with the anchors indexed by J deleted and $(p)_m$ the falling factorial: A^{p-k} times a bracket of degree k in p. A k-cut thus needs only k-connectedness while supplying a degree-k polynomial to tune. No counterexample above connectivity two is known: a search over thirty-eight branch families at $k = 3, 4, 5$ with up to 62 vertices, with (7) checked against brute force at $k = 3$, produced none, and it is not vacuous, those graphs carrying two to five gaps of total width 0.40 to 0.75 (`code/kcut.py`).

5.4 The Divisibility Lemma

The counterexamples are all instances of one algebraic fact, which we state once.

Lemma 13 (Divisibility)**.** *Let G be a finite graph, $S \subseteq V(G)$ with $|S| = k$, and suppose $G - S$ has p components, each inducing a copy of the same graph B. Write $A = \mu_B$. Then*

$$A^{p-k} \mid \mu_G.$$

Proof. Expand $\mu_G = \sum_M (-1)^{|M|} x^{n-2|M|}$ over matchings M of G and group the terms by

$$J(M) = \{\, i : \text{some edge of } M \text{ joins } S \text{ to the } i\text{-th component} \,\}.$$

A matching uses each vertex of S at most once, so distinct components in $J(M)$ are entered through distinct vertices of S; choosing one such vertex per component is an injection $J(M) \hookrightarrow S$, whence $|J(M)| \le k$. For $i \notin J(M)$ the restriction of M to the i-th component is an arbitrary matching of B, and summing over those choices independently contributes a factor $A^{p-|J(M)|}$. Each term of the resulting sum is therefore divisible by A^{p-k}, and hence so is μ_G. □

At $k = 2$ this is the divisor A^{p-2} of the two-cut identity. The exponent is positive exactly when $p > k$, which is why every construction below needs at least $k + 1$ branches. The counting step and the algebraic step are machine-checked separately (`RamaLean/DivisibilityLemma.lean`: `card_branches_le_sep`, `dvd_sum_of_pow_terms`, `matching_expansion_dvd`); the matching expansion itself is not, Mathlib carrying no matching polynomial.

5.5 Neither minimum degree nor connectivity

Connectivity is not the right invariant, because a graph of vertex connectivity κ has minimum degree at least κ, and it is the minimum degree that the counterexamples respond to. Every counterexample above has minimum degree at most two: Hall's has leaves, the 31-vertex graph of [17] has degree-2 vertices, and so do all three graphs of Proposition 12.

The two-hub construction separates the two conditions exactly, since it can be run with branches whose every vertex has degree at least three: ladders anchored at both ends of both rails, prisms, and $K_{3,q}$ blocks with two of the three degree-q vertices anchored. That yields 102 graphs of minimum degree three whose vertex connectivity is *two*, so each retains the separating pair the engine needs, and 77 of them have a gap of width at least 0.05, several wider than 2.9, against the 0.066 to 0.148 that the counterexamples of Proposition 12 occupy. None has a root of the bracket in a gap (`code/mindeg3.py`); that sweep did not test the roots of A, which is where the counterexample below lives.

Conjecture 14 (D3)**.** *Every finite graph of minimum degree at least three satisfies* $\mathrm{Zeros}(\mu_G) \subseteq \mathrm{spec}(T_G)$.

Conjecture 14 is false. We record the counterexample, which is smaller than Hall's by a factor of three.

Proposition 15 (D3 fails on 14 vertices)**.** *Let G have two hubs u, v, three copies of the star $K_{1,3}$ with centre c_i and leaves $\ell_{i,1}, \ell_{i,2}, \ell_{i,3}$, and every leaf joined to both hubs. Then G is simple, connected and bipartite on 14 vertices and 27 edges, with* $\deg c_i = \deg \ell_{i,j} = 3$ *and* $\deg u = \deg v = 9$*, so* $\delta(G) = 3$*; and*

$$\mu_G = x^4\,(x^2 - 3)\,(x^8 - 24x^6 + 171x^4 - 396x^2 + 270),$$

so $\sqrt{3}$ is a root of μ_G, while $\sqrt{3} \notin \mathrm{spec}(T_G)$.

Both halves are exact. That μ_G factors as displayed is computed from the graph by the deletion recursion, with $(x^2 - 3)$ divided out over $\mathbb{Q}$. That $\sqrt{3} \notin \mathrm{spec}(T_G)$ is Proposition 16, whose arithmetic is machine-checked (`RamaLean/D3Counterexample.lean`); the analytic input is the Angel–Friedman–Hoory criterion [13], which is cited and not reproved.

Proposition 16 (the certificate)**.** *On the four directed-edge orbits of G, write g_1 for centre-to-leaf, g_2 for leaf-to-centre, g_3 for leaf-to-hub and g_4 for hub-to-leaf. At $\lambda = \sqrt{3}$ the cavity system*

$$\lambda = \tfrac{1}{g_1} + 2g_3, \quad \lambda = \tfrac{1}{g_2} + 2g_1, \quad \lambda = \tfrac{1}{g_3} + 8g_4, \quad \lambda = \tfrac{1}{g_4} + g_2 + g_3$$

is solved by $g_1 = 1/\sqrt{3}$, $g_2 = \sqrt{3}$, $g_3 = 0$, $g_4 = \infty$, the last pair escaping along $g_3 g_4 \to -\frac{1}{8}$. The decay quotient has characteristic polynomial $t^4 - \frac{1}{8}t^2 - \frac{1}{2}$, whose largest root in modulus is

$$\rho = \sqrt{\tfrac{1+\sqrt{129}}{16}} = \tfrac{1}{4}\sqrt{1 + \sqrt{129}} = 0.878842\ldots < 1,$$

and the vertex Green's functions are $G_{\mathrm{hub}} = \sqrt{3}/3$, $G_{\mathrm{centre}} = -\sqrt{3}/6$, $G_{\mathrm{leaf}} = 0$, all finite.

Three things make this work where the naive attempt fails. The pole in g_4 is a pole of the *coordinate*: the decay quotient's entries diverge, but its characteristic polynomial does not, because a determinant collects only vertex-disjoint cycle collections and the two cycles present,

(g_3g_4) and $(g_1g_3g_4g_2)$, share vertices, so their weights $8(g_3g_4)^2 = \frac{1}{8}$ and $32g_1^2g_2^2(g_3g_4)^2 = \frac{1}{2}$ are finite and there is no cross term. Next, $\rho < 1$ needs no surd: with $s = t^2$ the polynomial is $8s^2 - s - 4$, and $(s-1)^2 \geq 0$ gives $8s^2 - 16s + 8 \geq 0$, which on substituting $8s^2 = s + 4$ leaves $15s \leq 12$, so $s \leq \frac{4}{5}$ (`decay_root_lt_one`). Last, the finiteness of every G_w excludes the one remaining possibility, that $\sqrt{3}$ is an *isolated point* of $\mathrm{spec}(T_G)$: an atom forces a pole of G_w with the eigenvector weight as residue, and there is none. That step is not a formality, since $\mathrm{spec}(T)$ is closed and the biregular cover of §4 carries exactly such an isolated $\{0\}$.

As an independent check, the exact $\sum_w G_w = \sqrt{3}/6$ is reproduced by the numerical resolvent to 7.7×10^{-10} at $\eta = 10^{-6}$, the error falling as η^2, and the exact ρ lies between the decays 0.892472 and 0.867159 computed at $\sqrt{3} \mp 0.005$ (`code/d3_exact.py`).

The mechanism is Hall's verbatim. His violating root $\sqrt{5}$ is the top matching root of the star $K_{1,5}$, whose matching polynomial is $x^4(x^2-5)$; the divisor is a star. A star's leaves have degree one *in the divisor* however large their degree is *in* G, so a hypothesis on $\delta(G)$ constrains the ambient graph and not the divisor. Here the branch is $K_{1,3}$, $A = \mu_{K_{1,3}} = x^2(x^2-3)$, and the two-cut identity $\mu_G = A^{p-2}\big(x^2A^2 - px(B_u + B_v)A + pDA + p(p-1)B_uB_v\big)$ makes every root of A a root of μ_G once $p \geq 3$. Three branches are needed and two do not suffice, since $A^0 = 1$ carries no root.

This is exactly what the searches above could not see, and the reason is worth stating because it is not a matter of scale. Each of them lifted the block's degrees to three by adding edges *inside* the block, which changes A and moves its roots; the counterexample lifts them from *outside*, by attaching each leaf to both hubs, which leaves A untouched. The five clean searches were therefore all run on blocks whose matching polynomial had already been perturbed away from the star. Separately, `code/mindeg3.py` iterated over the roots of the bracket and never over the roots of A, so the divisibility that is the whole engine went untested; that is repaired in `code/mindeg3adv.py`, which reports a signed margin rather than a verdict.

One trap on the way to Proposition 16 is worth recording, because it inverts the answer rather than blurring it, and because the same trap is what let five searches come back clean. Solved naively, the Angel–Friedman–Hoory system $\lambda = 1/r_e + \sum_{e \to f} r_f$ has four orbits here under $\mathrm{Aut}(G)$, and eliminating gives a quadratic whose two real branches both have decay above one, which reads as $\sqrt{3} \in \mathrm{spec}(T_G)$. That is wrong: a hair either side of $\sqrt{3}$ there are *three* real branches and the third has decay about 0.87. Its $r(\ell \to u)$ crosses zero exactly at $\sqrt{3}$, so the elimination multiplies by it and drops the branch, and a pointwise solver returns nothing there for the same reason. Proposition 16 is what the branch becomes once the singularity is removed rather than divided by.

Independently of the algebra, the density-of-states ladder settles the same point numerically and is what located it in the first place: with $g_e = 1/(z - \sum_{e \to f} g_f)$ at $z = \lambda + i\eta$, the quantity $|\operatorname{Im} \sum_w G_w|$ decays linearly in η outside the spectrum, tends to a positive constant inside a band, and diverges like η^{-1} at an atom. On this graph it reports an atom at 0, a band at 1.68, a band at 2.40, and linear decay at $\sqrt{3}$, the atom at the origin serving as the internal control (`code/d3_counterexample.py`).

Proposition 15 does not touch the connectivity statement, since G there has vertex connectivity two. But C2 falls to the same construction, and the reason it does is the point of this subsection.

Proposition 17 (C2 fails on 23 vertices)**.** *Let G have k hubs, p copies of $K_{1,m}$, and every* *leaf joined to* all k *hubs. A matching sends each hub into at most one branch, so at least* $p-k$ *branches contribute their full matching polynomial and*

$$A^{p-k} \mid \mu_G, \qquad A = \mu_{K_{1,m}} = x^{m-1}(x^2 - m),$$

which is the two-hub identity's A^{p-2} *for* $k = 2$*. At* $k = 3$*,* $m = 3$*,* $p = 5$ *the graph has* 23 *vertices, minimum degree three and vertex connectivity* three*, and* $\sqrt{3}$ *is a root of* μ_G *lying outside* $\operatorname{spec}(T_G)$ *(`code/c2_attack.py`).*

So neither stated hypothesis survives, and they fail for one reason rather than two: both constrain the ambient graph, while the engine is a property of the divisor. A star's leaves have degree one in H whatever their degree in G, and enlarging the separator leaves A untouched while raising κ.

We do not push the reading further, because the measurements do not support it. Raising the separator size k raises the divisibility to A^{p-k}, at a cost of $p \geq k+1$ branches, but it also raises every branch vertex's degree, and the two effects run against each other. Our scans resolve neither, so we make no claim about $\kappa \geq 4$ in either direction. What is established is that the two hypotheses actually proposed as repairs are false.

The implication D3 $\Rightarrow$ C2 (`MinimumDegreeThreshold.C2_of_D3`) and the sharpness statement (`threshold_is_three`) remain valid as implications, but both are now vacuous, their hypotheses being false.

5.6 What this leaves

Conjecture 10 is false, and so are both hypotheses proposed to repair it. The three refutations are one refutation: by Lemma 13 the branch polynomial A divides μ_G whenever a separator of size k is met by $p \geq k+1$ identical branches, and A depends on the branch alone. Minimum degree and vertex connectivity are conditions on G. Neither can see A, and a star divisor is exactly a branch whose own degrees stay at one however large the degrees of G become. Any repair of Conjecture 10 must constrain the divisor.

We do not know where that leaves the biregular case, which is the setting the phenomenon actually inhabits and is treated in [16]. There the cover is an (a, b)-biregular tree, the gap is the single interval $(0, |s-t|)$, and no construction of the present kind has reached it.

Two things are open and we state them as open rather than as programme. Whether some hypothesis on the *divisor* rescues the bound is untouched here; the natural candidate compares the divisor's Heilmann–Lieb radius $2\sqrt{\Delta_H - 1}$ with the band edge it would have to clear. And whether higher connectivity eventually defeats the construction is not settled by our measurements: raising the separator size k raises every branch vertex's degree as well, and the two effects run against each other. We looked, the picture did not resolve, and we make no claim in either direction.

6 Related work and formalization

The d-matching polynomial and the interval $[-\rho, \rho]$ are due to Hall, Puder and Sawin; the counterexample refuting the inclusion is due to Chris Hall [12] and is reproduced here with his permission, with our own independent verification of its certificate. The Angel–Friedman–Hoory ratio criterion [13] is what certifies the gap.

The Lean 4/Mathlib sources are in `RamaLean/` of the accompanying repository, and carry no `sorry` and no custom axioms. Formalized here: the closed form at first Betti number one and its corollaries, the counterexample's algebra, the branch and two-cut factorizations, the separation mechanism, and the minimum-degree threshold, including that minimum degree two does not suffice and that three is therefore exact. Statements are labelled `VERIFIED`, `HEURISTIC` or `CONJECTURE` in the sources, and the labels are used with their plain meaning: a `HEURISTIC` claim is supported by computation and is not proved.

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